

Supplementary Material to "Electronic properties governing the phase stability and elastic anisotropy of C14 and C15 Cr–Hf–Nb Laves phases"

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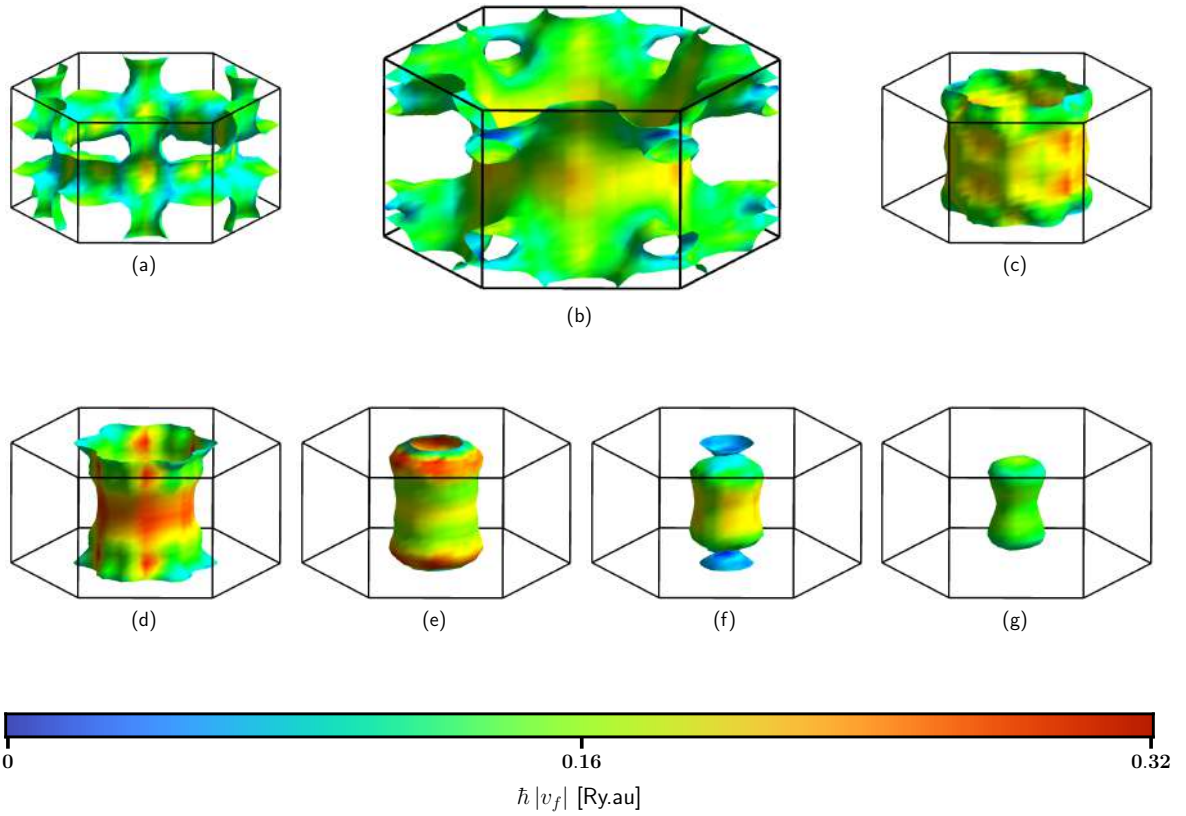


Figure 1 – HfNb₂-C14 seven independent sheets of the Fermi surface. The color map shows the magnitude of the Fermi velocity (1 $v_f \approx 2.188 \times 10^6$ m/s).

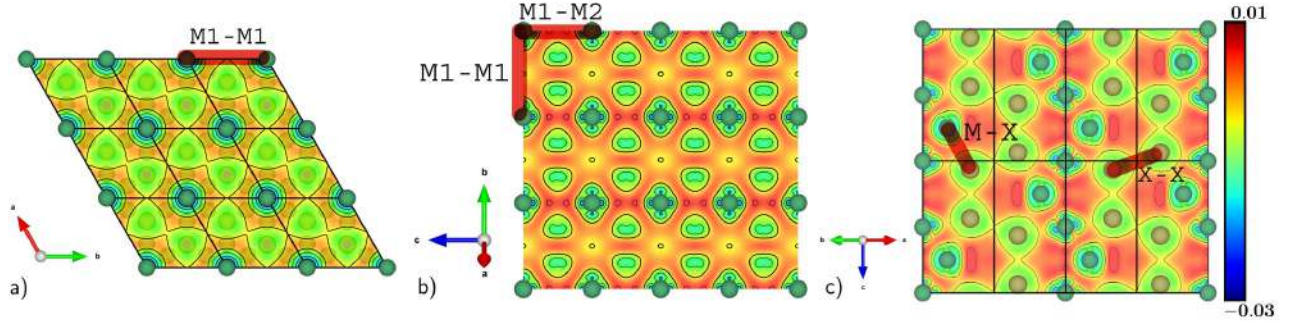


Figure 2 – Charge density minus superposition of atomic densities for HfNb₂-C14, GREEN - Nb, Yellow - Hf

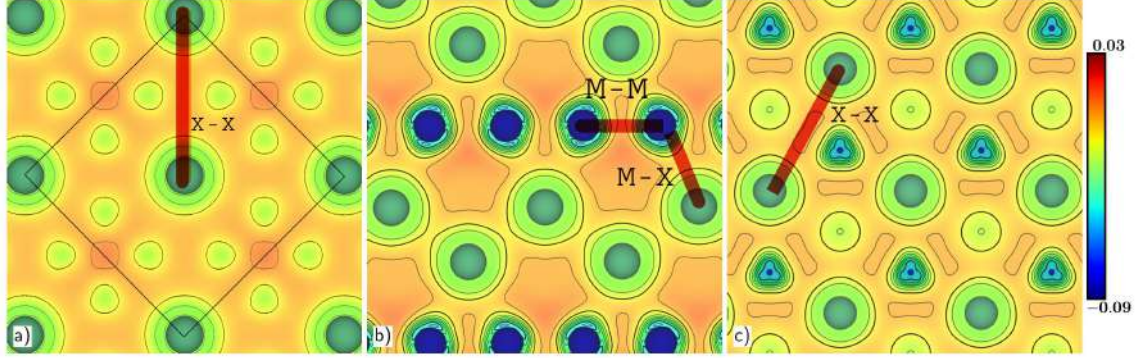


Figure 3 – Charge density minus superposition of atomic densities for NbCr₂-C15

Table 1 – Mulliken and Löwdin charges for distinct atomic sites in each C14 compound.

Compound	Site	Mulliken	Löwdin
HfCr ₂ -C14	Hf (4f)	+0.94	+0.31
	Cr ₁ (2a)	-0.42	-0.12
	Cr ₂ (6h)	-0.49	-0.17
NbCr ₂ -C14	Nb (4f)	+0.84	+0.15
	Cr ₁ (2a)	-0.43	-0.05
	Cr ₂ (6h)	-0.42	-0.08
HfNb ₂ -C14	Hf (4f)	+0.45	+0.45
	Nb ₁ (2a)	-0.30	-0.27
	Nb ₂ (6h)	-0.20	-0.21

Table 2 – Interatomic distances and ICOHP up to the Fermi level, in eV. Values for C14 XM₂ compounds.

Compound	M-X	M1-M2	M1-M1	X-X
HfCr ₂ -C14	2.896 Å / -1.140	4.046 Å / -0.093	5.048 Å / -0.013	3.068 Å / -1.395
NbCr ₂ -C14	2.858 Å / -1.105	4.035 Å / -0.104	4.897 Å / -0.022	3.073 Å / -1.197
HfNb ₂ -C14	3.193 Å / -1.183	4.442 Å / -0.231	5.556 Å / -0.012	3.308 Å / -1.144

Table 3 – Mulliken and Löwdin charges for C15 XM₂ compounds.

Compound	Mulliken (M, X)	Löwdin (M, X)
HfCr ₂ -C15	-0.48, +0.97	-0.16, +0.32
NbCr ₂ -C15	-0.45, +0.90	-0.09, +0.19
HfNb ₂ -C15	-0.26, +0.51	-0.26, +0.52

Table 4 – Interatomic distances and ICOHP up to the Fermi level, in eV. Values for C15 XM₂ compounds.

Compound	M-X	M-M	X-X
HfCr ₂ -C15	2.940 Å / -1.068	2.507 Å / -1.336	5.014 Å / -0.023
NbCr ₂ -C15	2.878 Å / -1.099	2.455 Å / -1.409	4.909 Å / -0.034
HfNb ₂ -C15	3.314 Å / -0.992	2.826 Å / -2.408	5.653 Å / -0.007

Table 9 – Minimum and maximum values of mechanical properties with anisotropy index shown alongside of plane (100) for C15.

Material	B (GPa)			G (GPa)			E (GPa)			ν		
	$B_{\max}$	$B_{\min}$	A^B	$G_{\max}$	$G_{\min}$	A^G	$E_{\max}$	$E_{\min}$	A^E	$\nu_{\max}$	$\nu_{\min}$	A^ν
HfCr ₂ -C15	190.96	190.96	0	71.934	64.072	0.032	186.640	172.879	0.027	0.377	0.297	0.051
NbCr ₂ -C15	229.197	229.197	0	76.345	61.097	0.062	195.183	168.332	0.052	0.438	0.278	0.093
HfNb ₂ -C15	113.467	113.467	0	22.734	19.801	0.038	61.788	56.137	0.034	0.459	0.359	0.052

Table 10 – Minimum and maximum values of mechanical properties with anisotropy index shown alongside of plane (110) for C15.

Material	B (GPa)			G (GPa)			E (GPa)			ν		
	$B_{\max}$	$B_{\min}$	A^B	$G_{\max}$	$G_{\min}$	A^G	$E_{\max}$	$E_{\min}$	A^E	$\nu_{\max}$	$\nu_{\min}$	A^ν
HfCr ₂ -C15	190.959	190.959	0	71.934	64.072	0.032	191.727	172.879	0.034	0.377	0.297	0.042
NbCr ₂ -C15	229.197	229.197	0	76.345	61.097	0.062	206.144	168.332	0.066	0.438	0.278	0.077
HfNb ₂ -C15	113.467	113.467	0	22.734	19.801	0.038	63.933	56.137	0.042	0.459	0.359	0.041

Table 11 – Minimum and maximum values of mechanical properties with anisotropy index shown alongside of plane (111) for C15.

Material	B (GPa)			G (GPa)			E (GPa)			ν		
	$B_{\max}$	$B_{\min}$	A^B	$G_{\max}$	$G_{\min}$	A^G	$E_{\max}$	$E_{\min}$	A^E	$\nu_{\max}$	$\nu_{\min}$	A^ν
HfCr ₂ -C15	190.959	190.959	0	71.934	64.071	0.031	186.640	186.640	0	0.377	0.297	0.062
NbCr ₂ -C15	229.197	229.197	0	76.345	61.096	0.059	195.184	195.184	0	0.438	0.278	0.118
HfNb ₂ -C15	113.467	113.467	0	22.734	19.801	0.036	61.788	61.788	0	0.460	0.359	0.065

Table 12 – Minimum and maximum values of mechanical properties with anisotropy index shown alongside of plane (001) for C14.

Material	B (GPa)			G (GPa)			E (GPa)			ν		
	$B_{\max}$	$B_{\min}$	A^B	$G_{\max}$	$G_{\min}$	A^G	$E_{\max}$	$E_{\min}$	A^E	$\nu_{\max}$	$\nu_{\min}$	A^ν
HfCr ₂ -C14	197.027	197.027	0	86.252	53.853	0.167	217.865	217.865	0	0.368	0.263	0.118
NbCr ₂ -C14	228.010	228.010	0	80.089	59.124	0.108	217.184	217.184	0	0.356	0.327	0.030
HfNb ₂ -C14	146.009	146.009	0	39.773	33.877	0.057	113.386	113.386	0	0.425	0.316	0.105

Table 13 – Minimum and maximum values of mechanical properties with anisotropy index shown alongside of plane (100) for C14.

Material	B (GPa)			G (GPa)			E (GPa)			ν		
	$B_{\max}$	$B_{\min}$	A^B	$G_{\max}$	$G_{\min}$	A^G	$E_{\max}$	$E_{\min}$	A^E	$\nu_{\max}$	$\nu_{\min}$	A^ν
HfCr ₂ -C14	197.027	179.388	0.033	86.252	53.853	0.125	217.865	159.387	0.099	0.486	0.228	0.189
NbCr ₂ -C14	231.056	228.010	0.005	82.902	59.124	0.097	224.699	174.175	0.085	0.473	0.273	0.128
HfNb ₂ -C14	146.009	144.646	0.003	44.919	33.877	0.057	127.005	98.123	0.081	0.453	0.316	0.087

Table 14 – Minimum and maximum values of mechanical properties with anisotropy index shown alongside of plane ($\bar{1}\bar{1}0$) and C14.

Material	B (GPa)			G (GPa)			E (GPa)			ν		
	$B_{\max}$	$B_{\min}$	A^B	$G_{\max}$	$G_{\min}$	A^G	$E_{\max}$	$E_{\min}$	A^E	$\nu_{\max}$	$\nu_{\min}$	A^ν
HfCr ₂ -C14	197.026	179.388	0.033	86.252	53.853	0.125	217.861	159.386	0.099	0.486	0.228	0.189
NbCr ₂ -C14	231.056	228.010	0.005	82.902	59.124	0.097	224.703	174.175	0.085	0.473	0.273	0.128
HfNb ₂ -C14	146.009	144.646	0.003	44.919	33.877	0.074	127.007	98.122	0.081	0.454	0.316	0.087

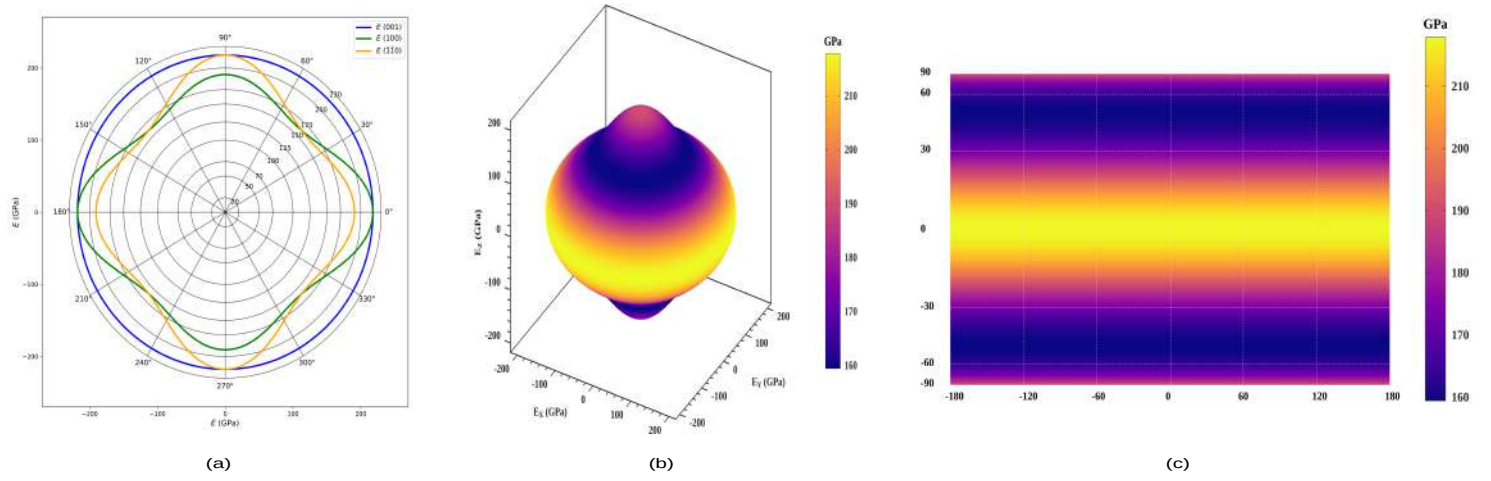


Figure 4 – Graphical representations of the Young's modulus (E) anisotropy for the HfCr₂-C14 system: (a) 2D polar plots for (001), (100), and $(\bar{1}\bar{1}0)$ planes; (b) 3D surface visualization; (c) Gall-Peters map projection view.

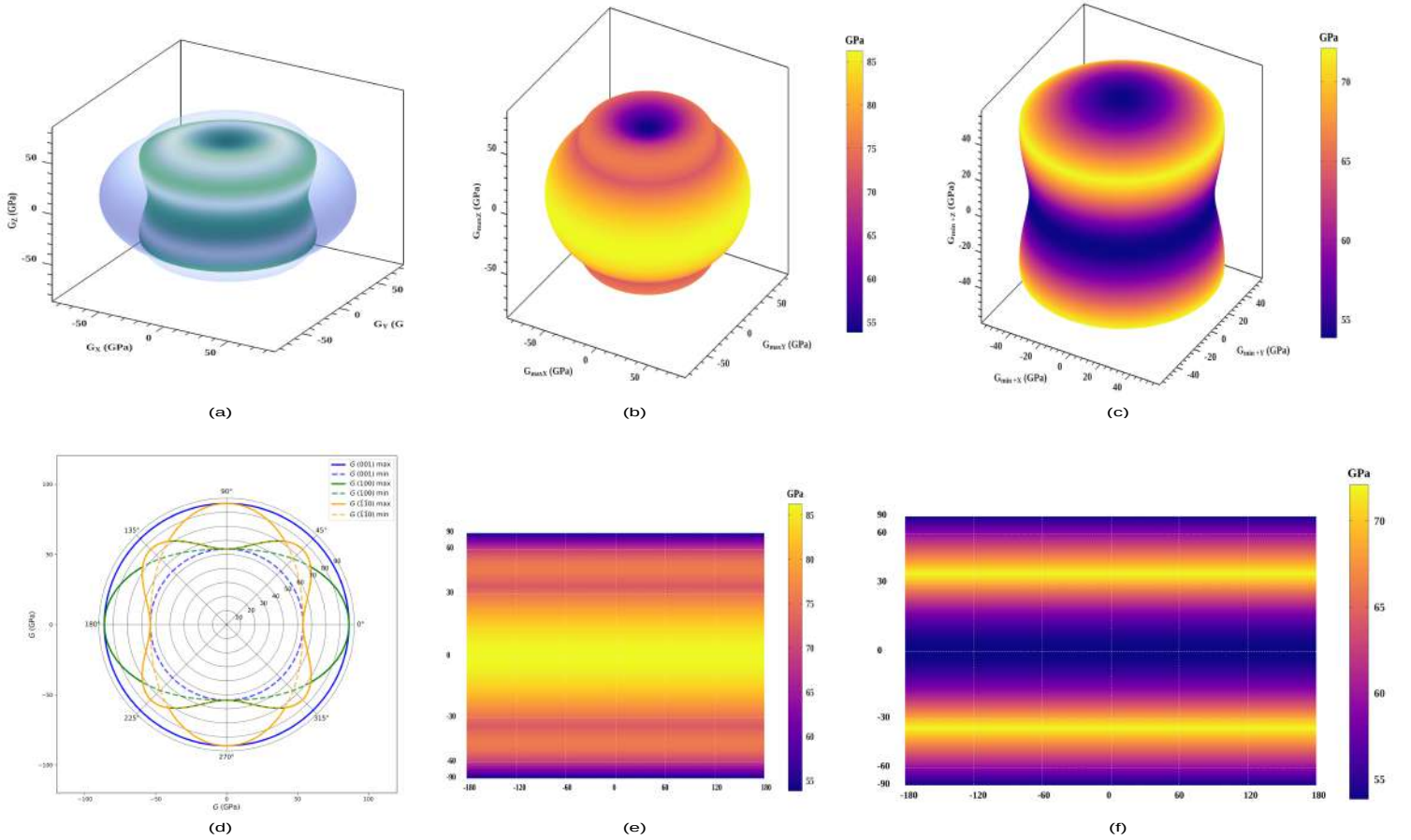


Figure 5 – Graphical representations of the Shear modulus (G) anisotropy for the HfCr₂-C14 system: (a) 3D surface visualization; (b), (c) 3D surface visualization G_{max} and G_{min} ; (d) 2D polar plots for (001), (100), and $(\bar{1}\bar{1}0)$ planes; (e), (f) Gall-Peters map projection G_{max} and G_{min} .

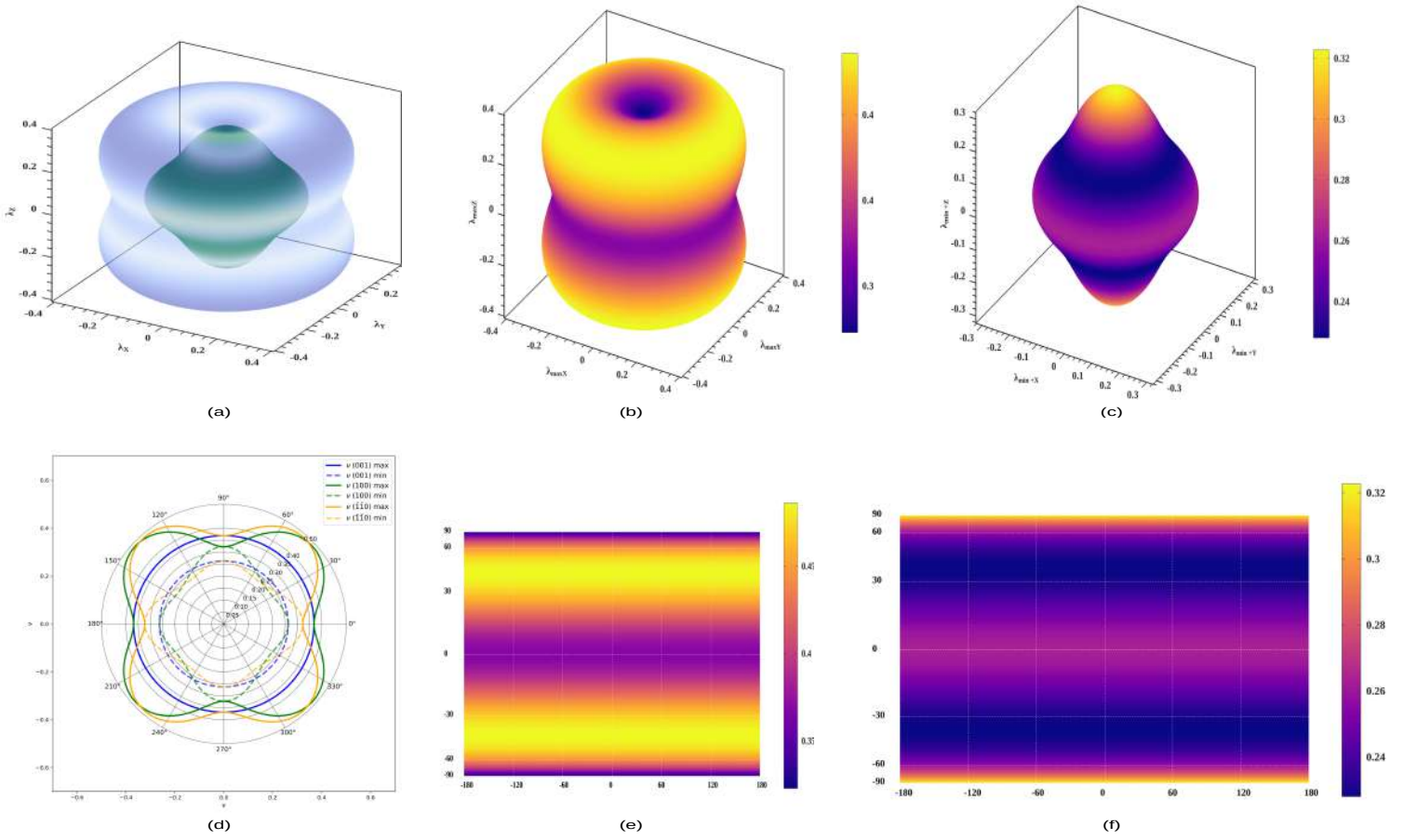


Figure 6 – Graphical representations of the Poisson's modulus (ν) anisotropy for the HfCr₂-C14 system: (a) 3D surface visualization; (b),(c) 3D surface visualization $\nu_{\max}$ and $\nu_{\min}$; (d) 2D polar plots for (001), (100), and ($\bar{1}\bar{1}0$) planes; (e),(f) Gall-Peters map projection $\nu_{\max}$ and $\nu_{\min}$.

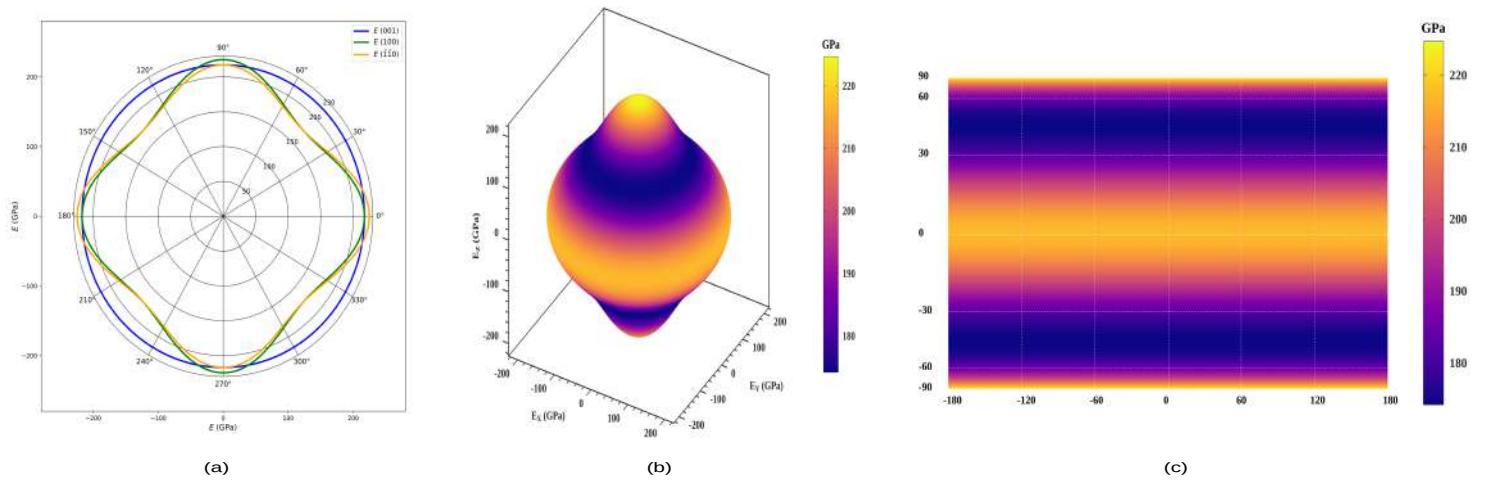


Figure 7 – Graphical representations of the Young's modulus (E) anisotropy for the NbCr₂-C14 system: (a) 2D polar plots for (001), (100), and ($\bar{1}\bar{1}0$) planes; (b) 3D surface visualization; (c) Gall-Peters map projection view.

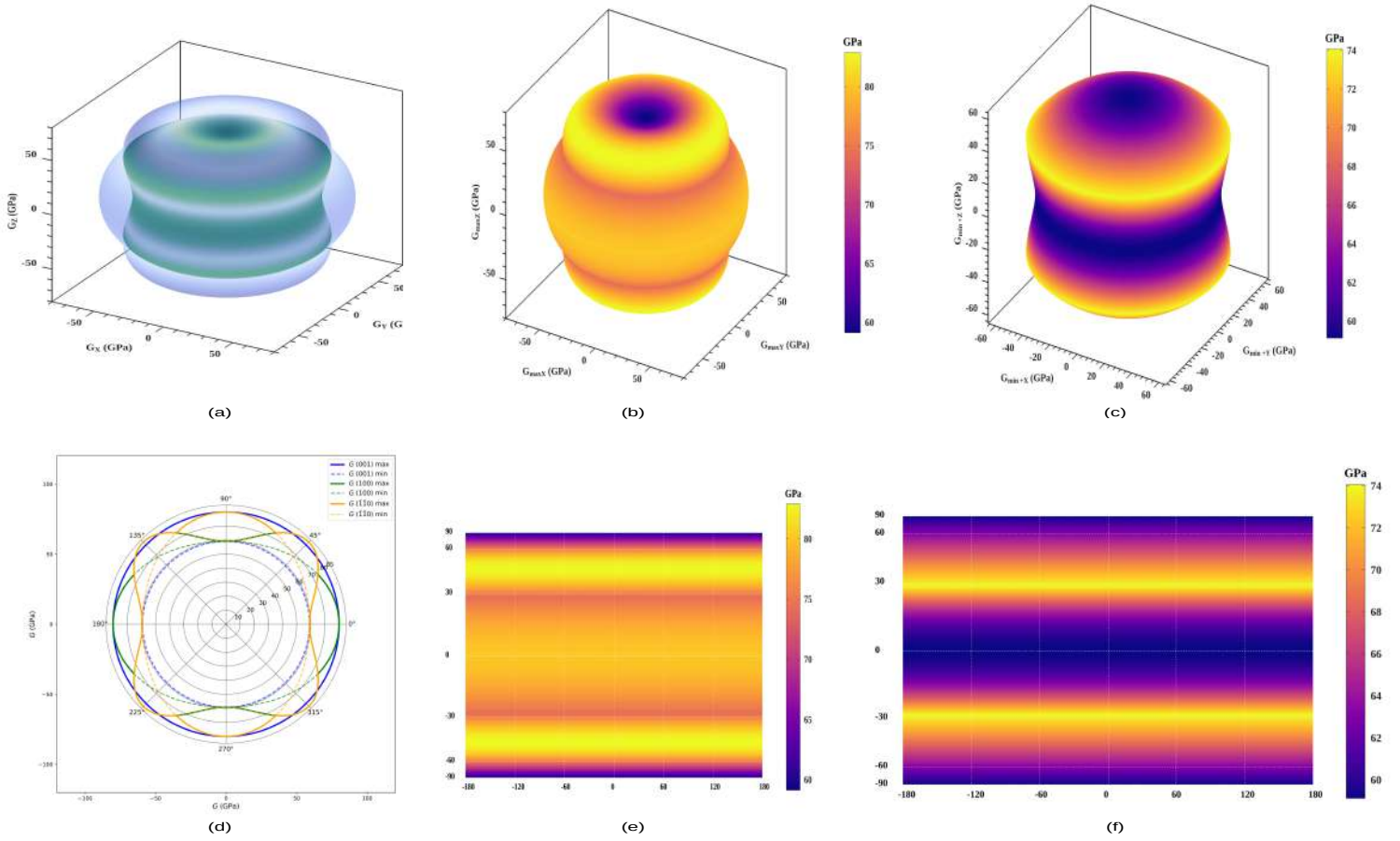


Figure 8 – Graphical representations of the Shear modulus (G) anisotropy for the NbCr₂-C14 system: (a) 3D surface visualization; (b),(c) 3D surface visualization $G_{\max}$ and $G_{\min}$; (d) 2D polar plots for (001), (100), and ($\bar{1}\bar{1}0$) planes; (e),(f) Gall-Peters map projection $G_{\max}$ and $G_{\min}$.

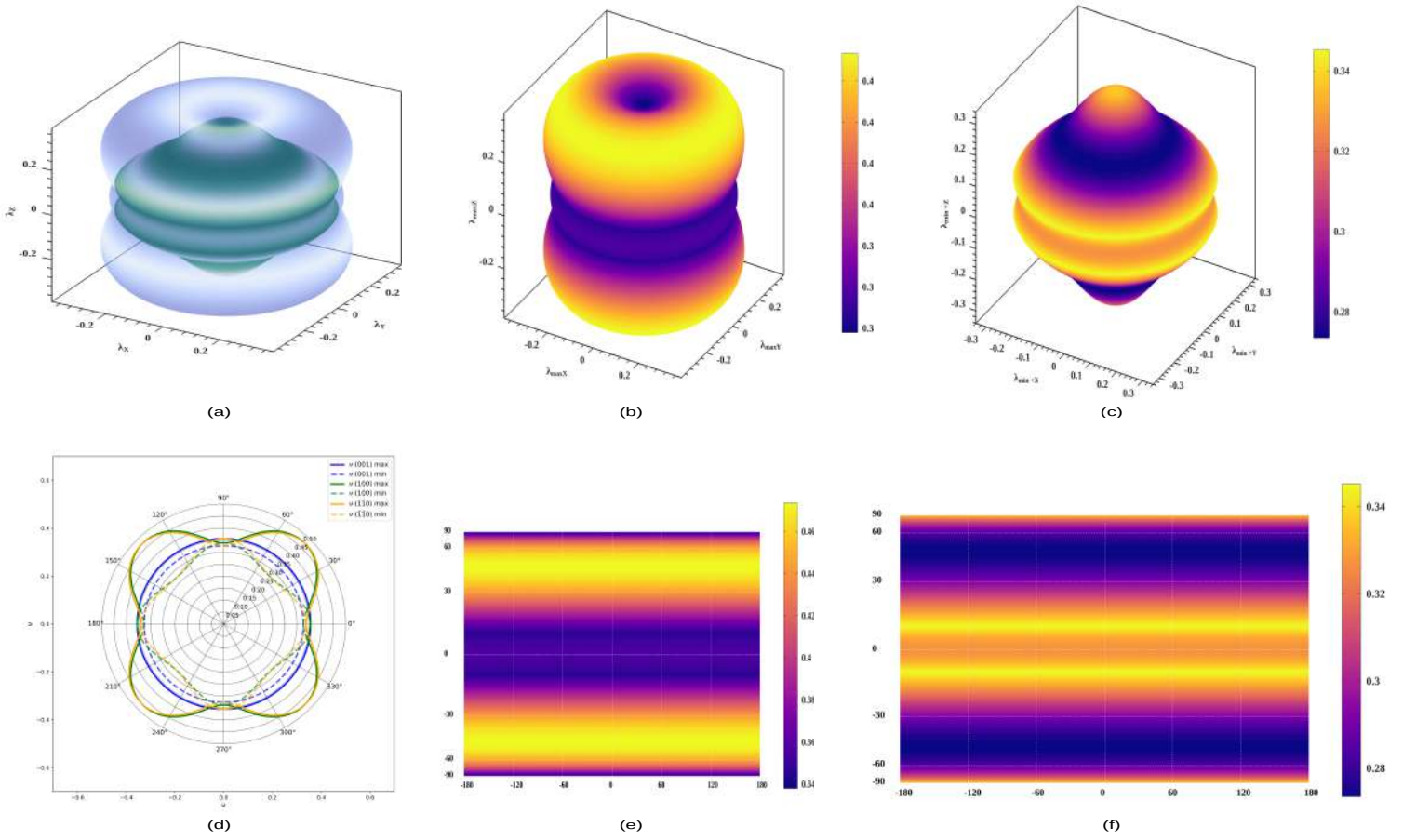


Figure 9 – Graphical representations of the Poisson's modulus (ν) anisotropy for the NbCr₂-C14 system: (a) 3D surface visualization; (b),(c) 3D surface visualization $\nu_{\max}$ and $\nu_{\min}$; (d) 2D polar plots for (001), (100), and ($\bar{1}\bar{1}0$) planes; (e),(f) Gall-Peters map projection $\nu_{\max}$ and $\nu_{\min}$.

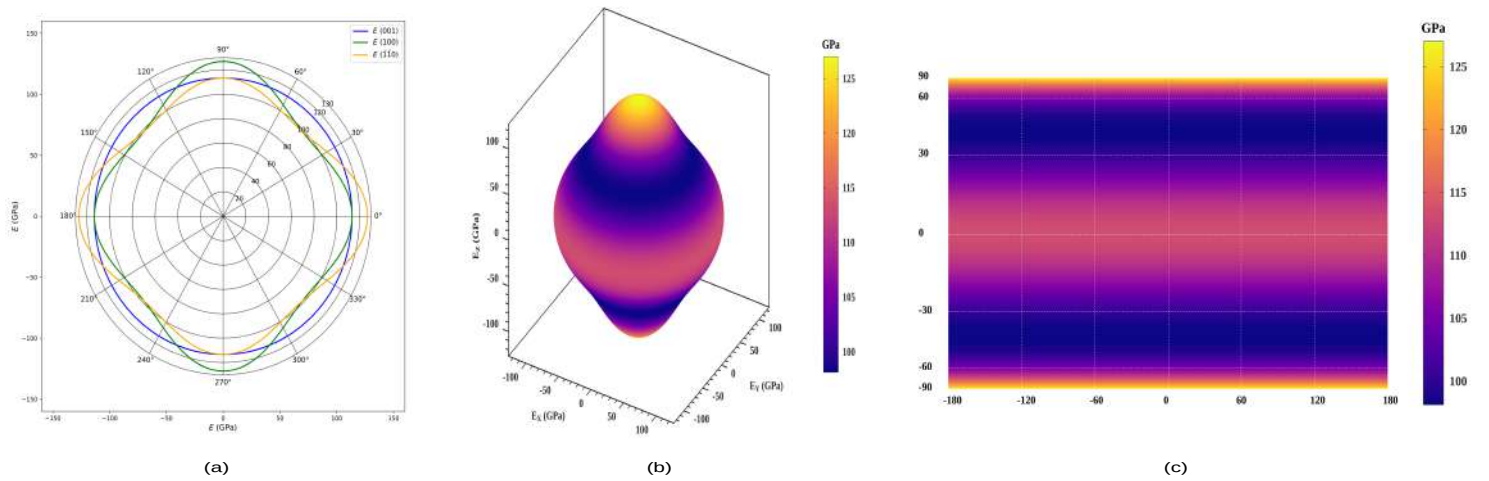


Figure 10 – Graphical representations of the Young's modulus (E) anisotropy for the HfNb₂-C14 system: (a) 2D polar plots for (001), (100), and ($\bar{1}\bar{1}0$) planes; (b) 3D surface visualization; (c) Gall-Peters map projection view.

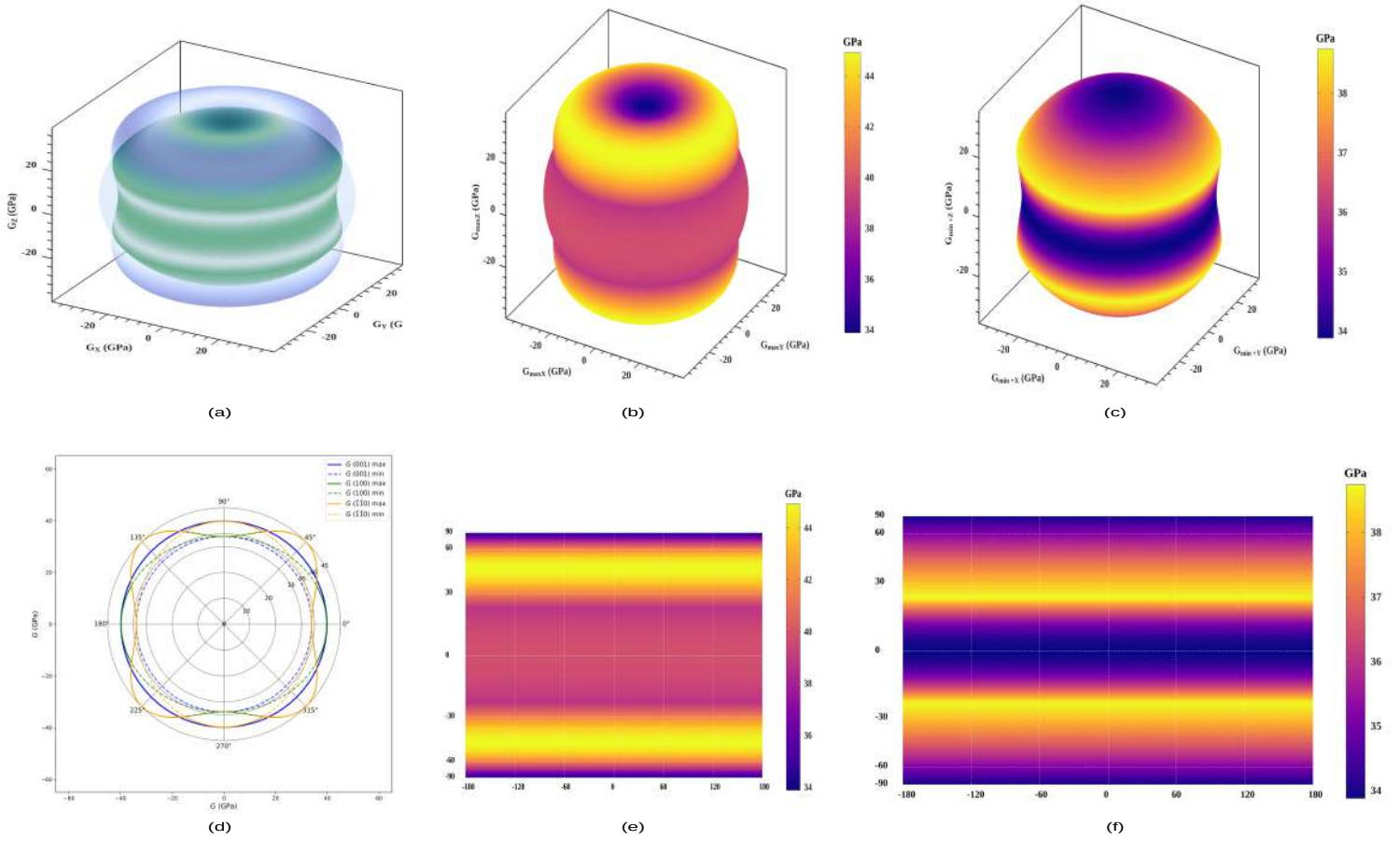


Figure 11 – Graphical representations of the Shear modulus (G) anisotropy for the HfNb₂-C14 system: (a) 3D surface visualization; (b),(c) 3D surface visualization G_{max} and G_{min} ; (d) 2D polar plots for (001), (100), and ($\bar{1}\bar{1}0$) planes; (e),(f) Gall-Peters map projection G_{max} and G_{min} .

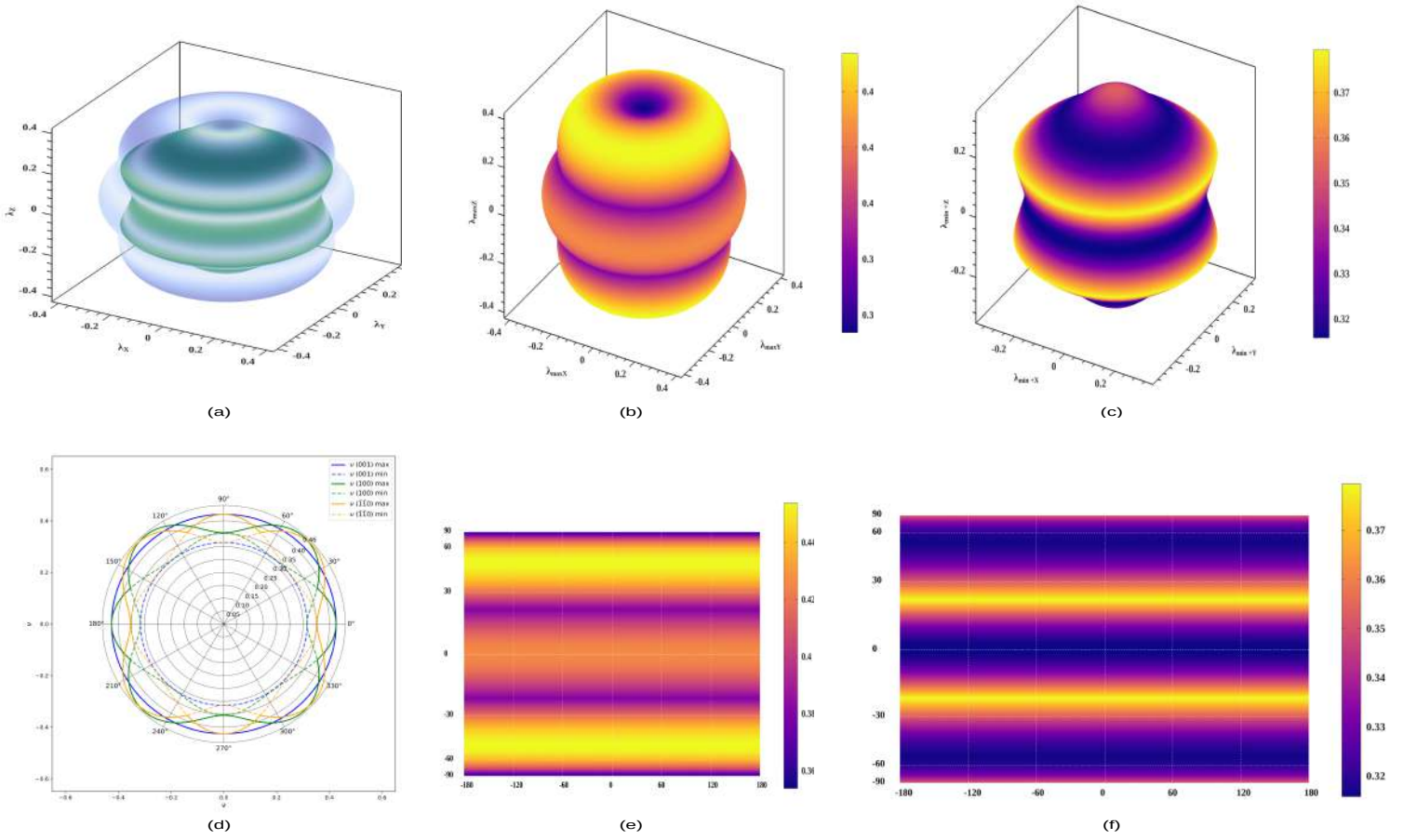


Figure 12 – Graphical representations of the Poisson's modulus (ν) anisotropy for the HfNb₂-C14 system: (a) 3D surface visualization; (b),(c) 3D surface visualization $\nu_{\max}$ and $\nu_{\min}$; (d) 2D polar plots for (001), (100), and ($\bar{1}\bar{1}0$) planes; (e),(f) Gall-Peters map projection $\nu_{\max}$ and $\nu_{\min}$.

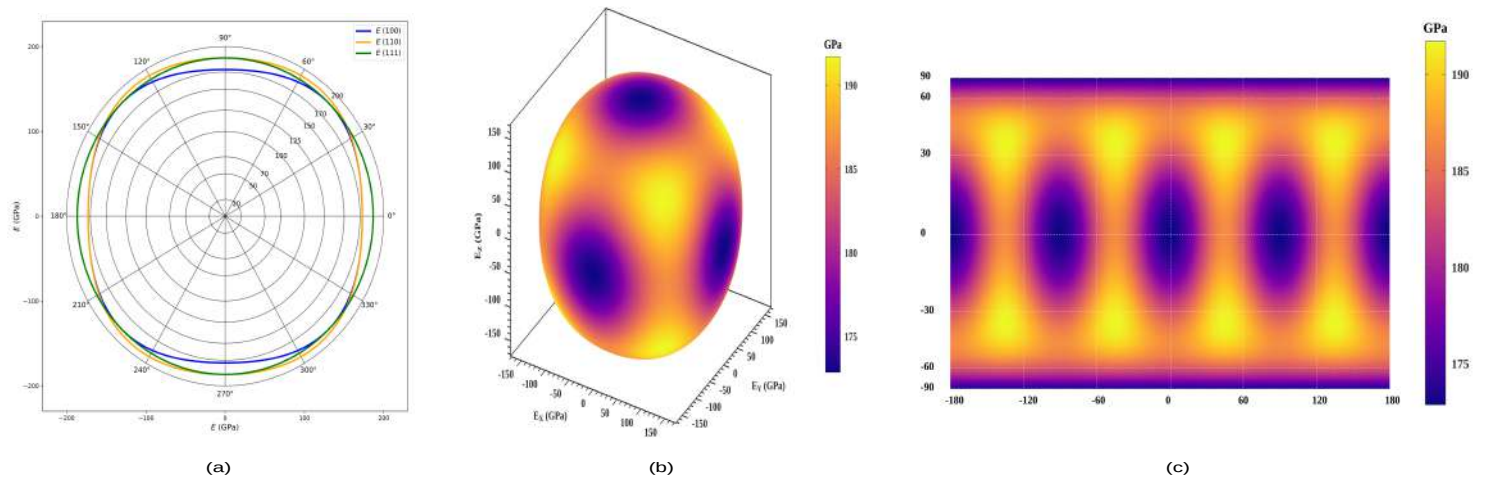


Figure 13 – Graphical representations of the Young's modulus (E) anisotropy for the HfCr₂-C15 system: (a) 2D polar plots for (001), (110), and (111) planes; (b) 3D surface visualization; (c) Gall-Peters map projection view.

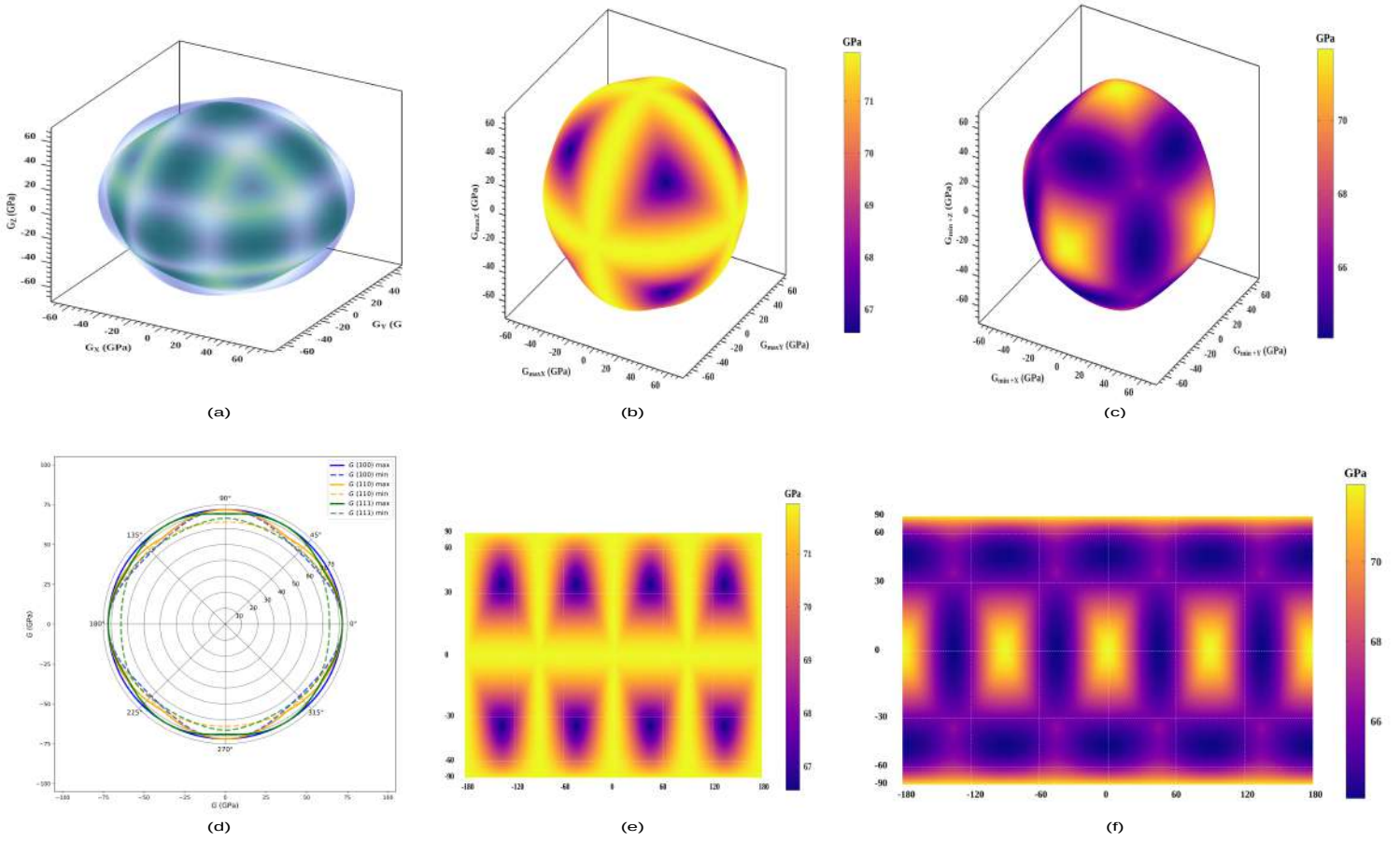


Figure 14 – Graphical representations of the Shear modulus (G) anisotropy for the $\text{HfCr}_2\text{-C15}$ system: (a) 3D surface visualization; (b),(c) 3D surface visualization G_{max} and G_{min} ; (d) 2D polar plots for (001), (110), and (111) planes; (e),(f) Gall-Peters map projection G_{max} and G_{min} .

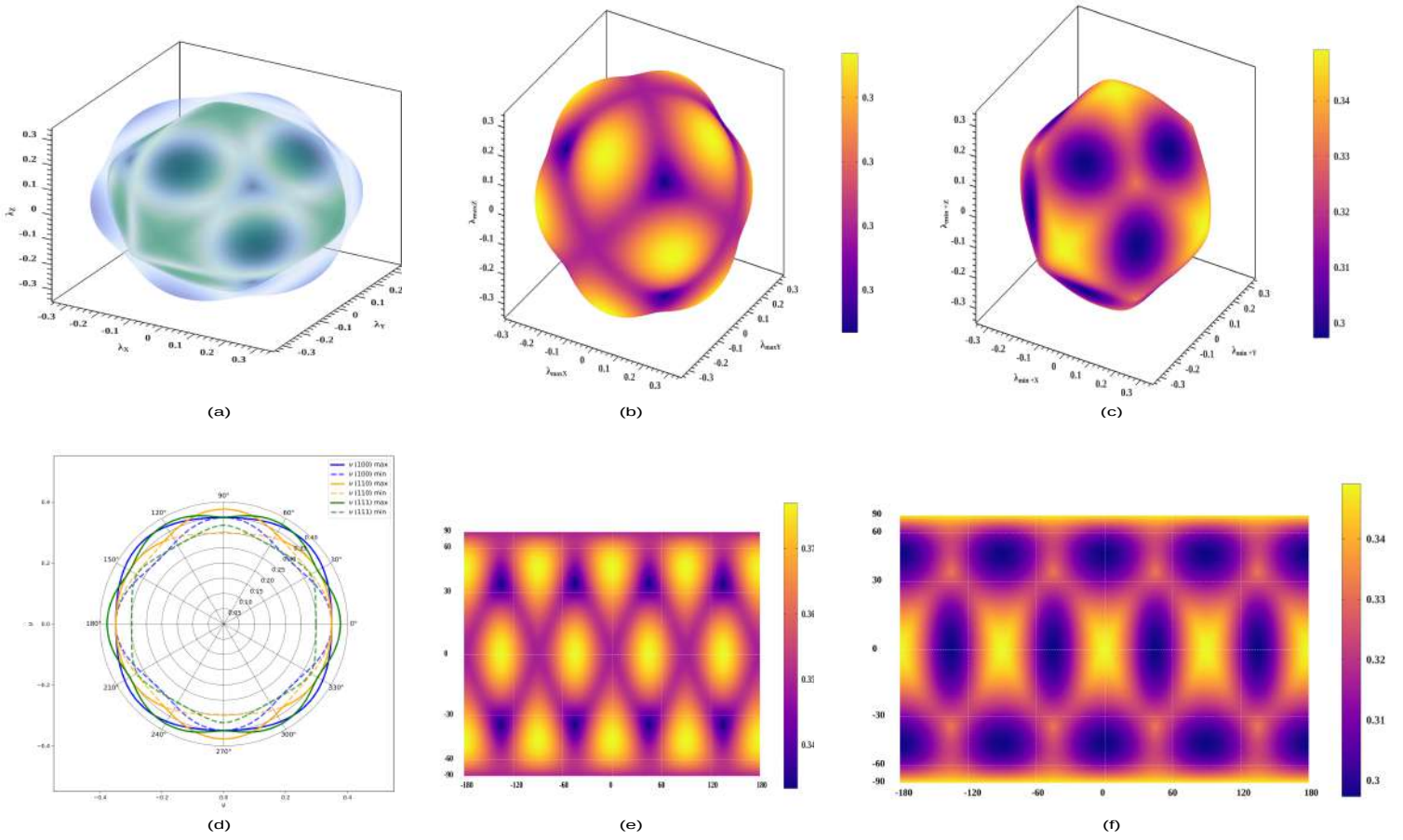


Figure 15 – Graphical representations of the Poisson's modulus (ν) anisotropy for the $\text{HfCr}_2\text{-C15}$ system: (a) 3D surface visualization; (b),(c) 3D surface visualization ν_{max} and ν_{min} ; (d) 2D polar plots for (001), (110), and (111) planes; (e),(f) Gall-Peters map projection ν_{max} and ν_{min} .

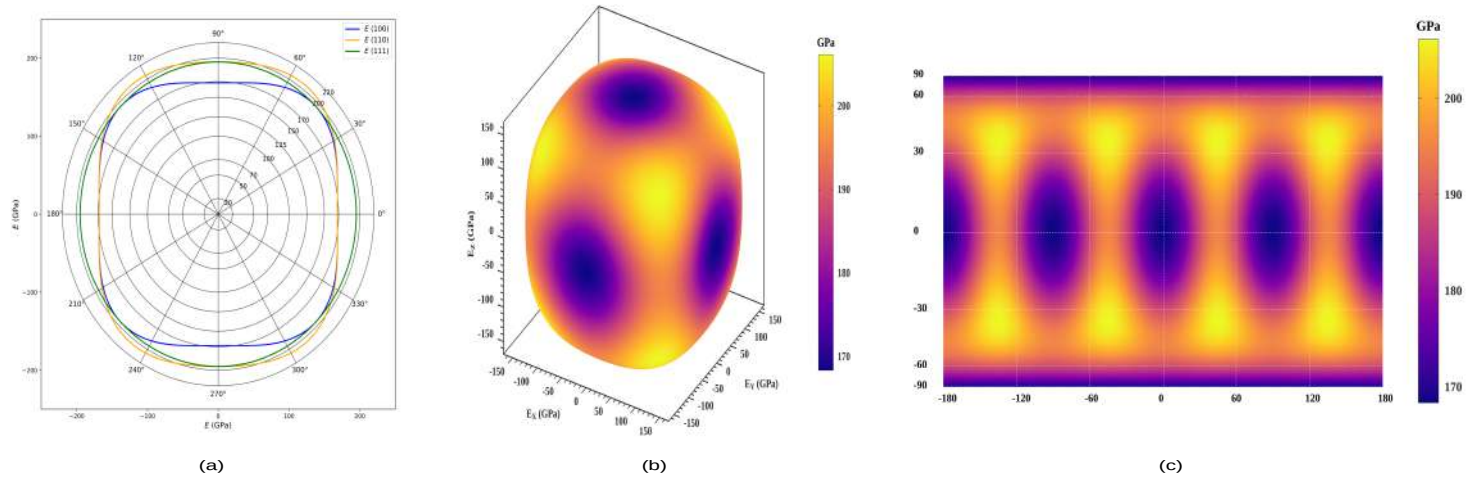


Figure 16 – Graphical representations of the Young's modulus (E) anisotropy for the $\text{NbCr}_2\text{-C15}$ system:(a) 2D polar plots for (001), (110), and (111) planes; (b) 3D surface visualization; (c) Gall-Peters map projection view..

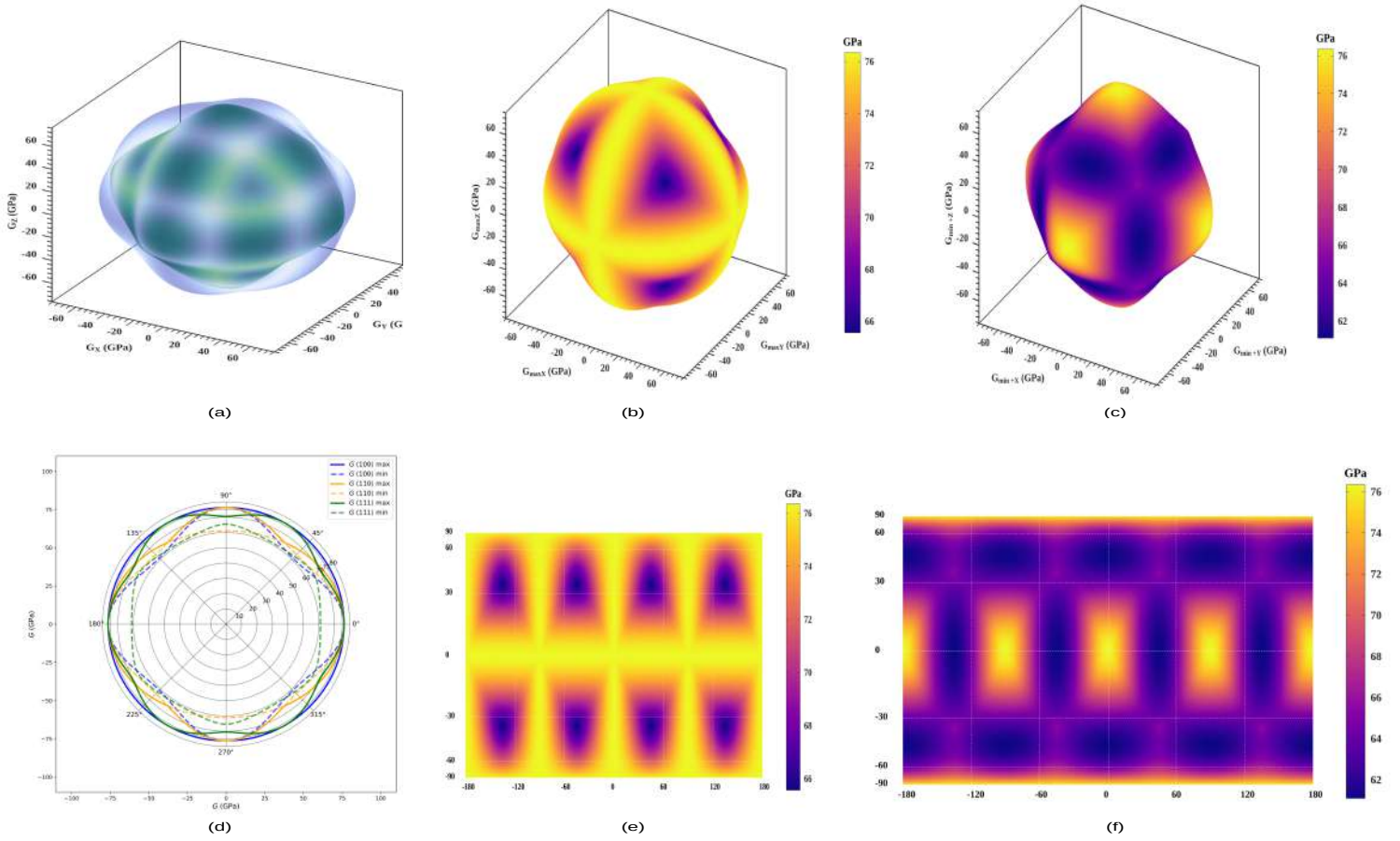


Figure 17 – Graphical representations of the Shear modulus (G) anisotropy for the NbCr₂-C15 system: (a) 3D surface visualization; (b),(c) 3D surface visualization $G_{\max}$ and $G_{\min}$; (d) 2D polar plots for (001), (110), and (111) planes; (e),(f) Gall-Peters map projection $G_{\max}$ and $G_{\min}$.

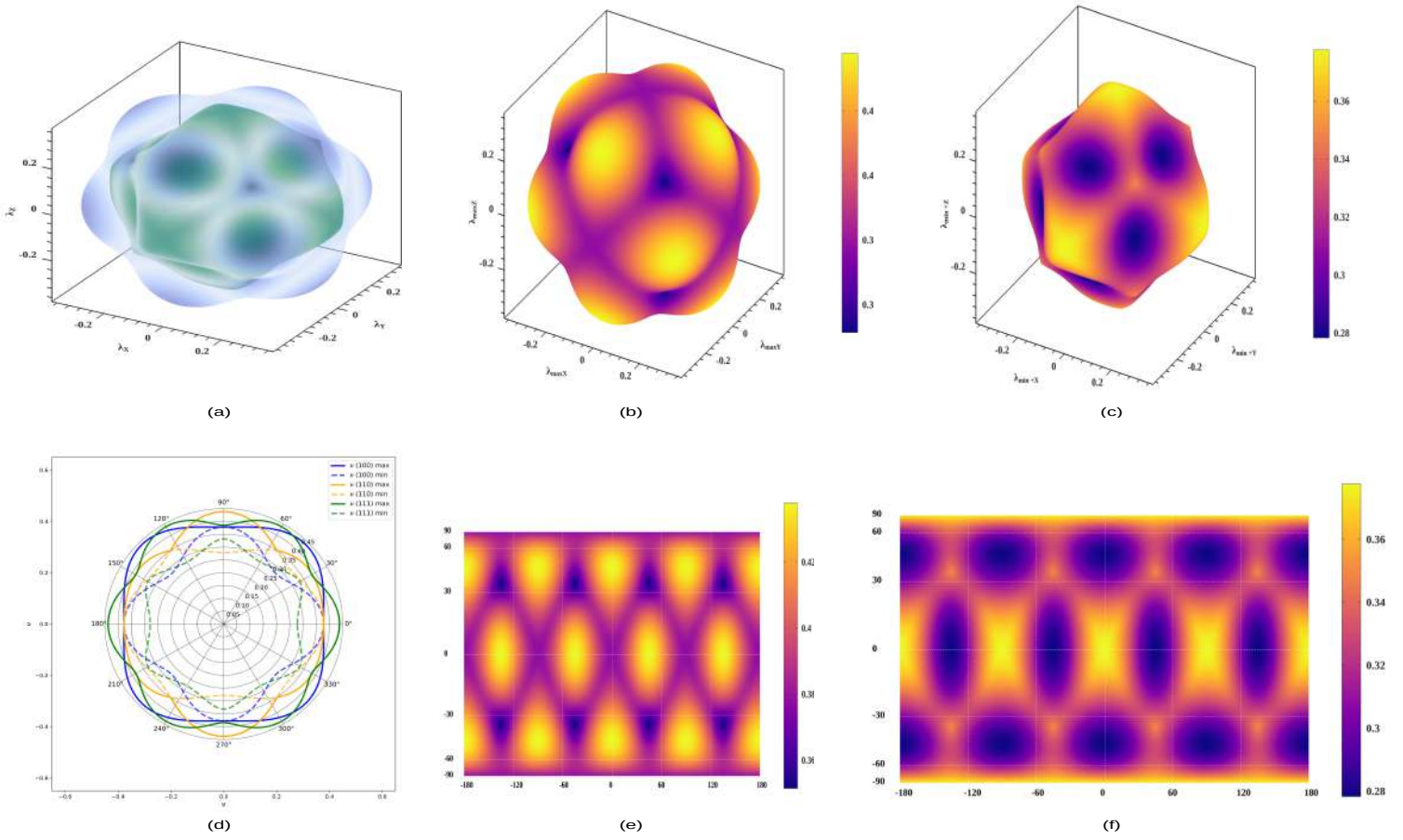


Figure 18 – Graphical representations of the Poisson's modulus (ν) anisotropy for the NbCr₂-C15 system: (a) 3D surface visualization; (b),(c) 3D surface visualization $\nu_{\max}$ and $\nu_{\min}$; (d) 2D polar plots for (001), (110), and (111) planes; (e),(f) Gall-Peters map projection $\nu_{\max}$ and $\nu_{\min}$.

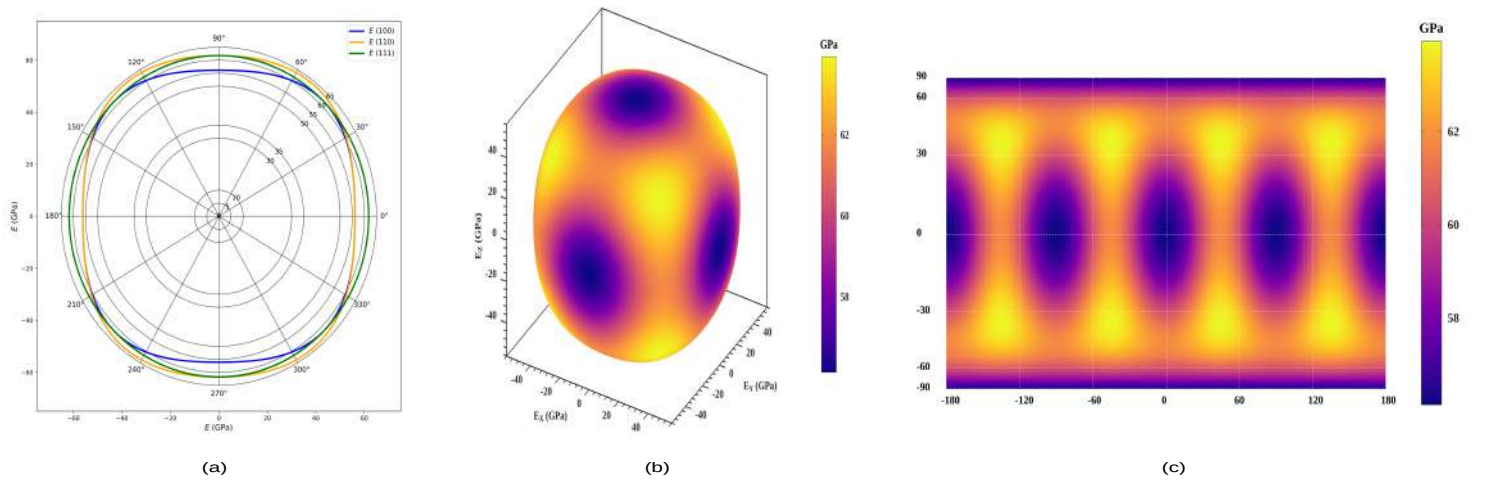


Figure 19 – Graphical representations of the Young's modulus (E) anisotropy for the HfNb₂-C15 system: (a) 2D polar plots for (001), (110), and (111) planes; (b) 3D surface visualization; (c) Gall-Peters map projection view..

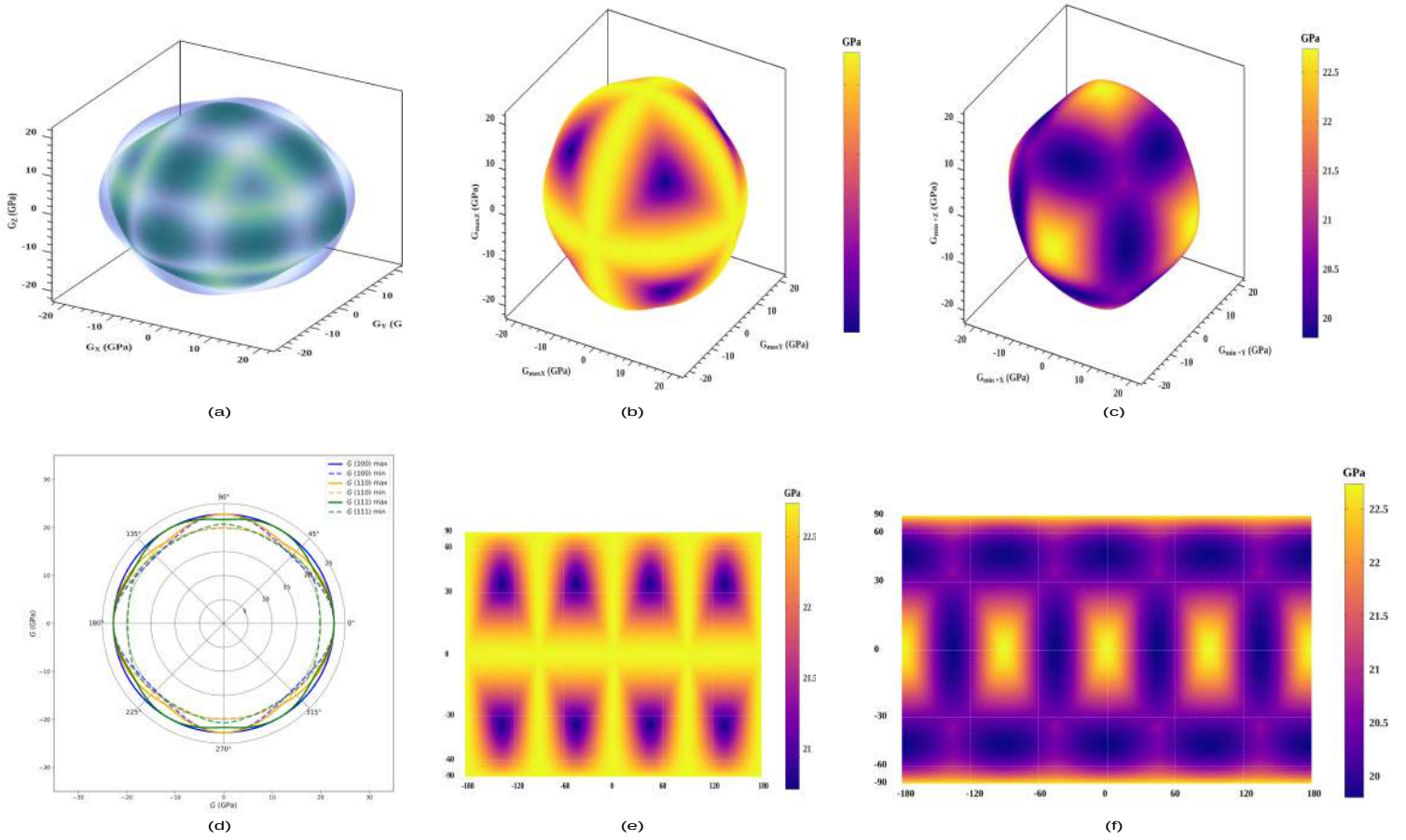


Figure 20 – Graphical representations of the Shear modulus (G) anisotropy for the $\text{HfNb}_2\text{-C15}$ system: (a) 3D surface visualization; (b),(c) 3D surface visualization G_{max} and G_{min} ; (d) 2D polar plots for (001), (110), and (111) planes; (e),(f) Gall-Peters map projection G_{max} and G_{min} .

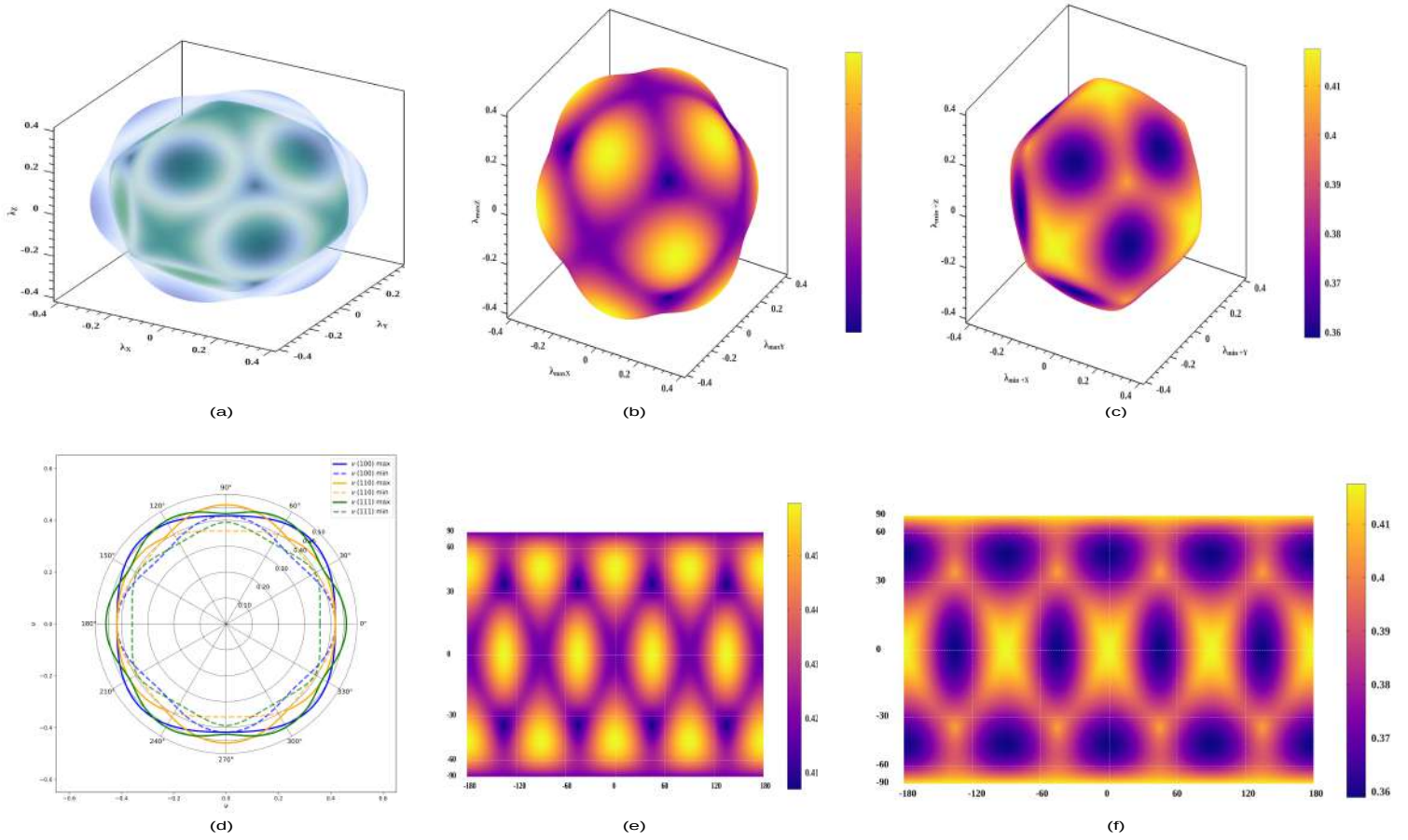


Figure 21 – Graphical representations of the Poisson's modulus (ν) anisotropy for the HfNb₂-C15 system: (a) 3D surface visualization; (b),(c) 3D surface visualization $\nu_{\max}$ and $\nu_{\min}$; (d) 2D polar plots for (001), (110), and (111) planes; (e),(f) Gall-Peters map projection $\nu_{\max}$ and $\nu_{\min}$.